Experiment - 2

AIM

Convert the given ER diagram into relational tables using MySQL with Primary Key, foreign Key, Not Null, UNIQUE and on DELETE CASCADE.

Source Code

```
Create Database E-Commerce;
USE E-Commerce;

Create Table Customer (
Customer_id int primary key,
name VARCHAR(100) Not Null,
email VARCHAR(100) UNIQUE,
Phone_no VARCHAR(15),
dob Data
);

Create table Seller (
  Seller_id int primary key,
  Seller_name VARCHAR(100) Not Null,
  email VARCHAR(100) UNIQUE,
  Phone_no VARCHAR(15)
);

Create table Category (
 Category_id int primary key,
 Category_name VARCHAR(100) Not Null
UNIQUE
);
```

```
Create table Address (
address_id int primary key,
State VARCHAR (50),
house_no VARCHAR (20),
Street VARCHAR (150)
City VARCHAR (50),
Pincode VARCHAR (10)
);

Create table product (
Product_id int primary key.
Product_name VARCHAR (100) Not Null,
Price DECIMAL (10,2) Not Null,
Stock INT,
Category_id INT,
Seller_id INT,
Foreign KEY (Category_id) Referenes
Category (category_id)
ON DELETE CASCADE,
FOREIGN KEY (Seller_id) Referenes
Seller (Seller_id)
ON DELETE CASCADE
);

Create table Payment (
Payment_id INT PRIMARY KEY,
Payment_date DATE,
amount DECIMAL (10,2) Not Null,
Payment_Method VARCHAR (50),
Payment_Status VARCHAR (50)
);
```

```
Create table Orders (
 Order_id INT PRIMARY KEY,
 Order_date DATE,
 total_amount DECIMAL (10,2),
 Customer_id INT,
 address_id INT,
Payment_id INT,
delivery_id INT,
foreign KEY (Customer_id) References
Customer (Customer_id)
 ON DELETE CASCADE,
 foreign key (address_id) References
 address (address_id)
 ON DELETE CASCADE,
 foreign key (delivery_id) References
 delivery (delivery_id)
  ON DELETE CASCADE
 );

 Create table Order_item (
 Order_id INT,
Product_id INT,
 Quantity INT (Not Null),
PRIMARY KEY (Order_id, Product_id),
 foreign key (Order_id) References
Product (Product_id),
  ON DELETE CASCADE
);
```

```
Create Table Order_item (
Order_id INT,
PRODUCT_ID INT,
Quantity INT (Not Null),
PRIMARY KEY (Order_id, Product_id);
foreign key (Order_id) Referens
Product (Product_id)
 ON DEIETE CASCADE
);
```

SAMPLE DATA →

```
INSERT INTO Customer Values
(1, 'Rahul
Sharma', 'rahul@gmail.com', '9876543210'
 , '2004-05-10'),
(2, 'Amrita
Varma', 'amrita@gmail.com', '6392927272 11'
 , '2003-08-15'),
(3, 'Priya
Sigh ', 'Priya@gmail.com', ' 7272637293'
 , '2009-06-16');

INSERT INTO seller Values
(1, 'Tech
World', 'tech@gmail.Com', '9811117898'),
(2, 'fashion
hub', 'fashion@gmail.com', ' 639272721 1');

INSERT INTO Category Values
(1, 'Electronics'),
(2, 'Clothing'),
(3, 'Books');
```

```
INSERT INTO Product Values
(101, 'Laptop', 55000, 10, 1, 1),
(102, 'T-Shirt', 799, 25, 2, 2);

INSERT INTO Payment Values
(201, '2026-09-01', 55000, 'UPI', 'Completed'),
(202, '206-09-02', 799, 'Card', 'Completed');

INSERT INTO Address Values
(301, 'haryana', '12', 'MG Road', 'Gurgaon', '122001'),
(302, 'Maharashtra', '25', 'Link Road', 'Mumbai', '400001');

INSERT INTO Delivery Values
(301, 'TRK1001', '2026-09-03'),
(302, 'TRK1002', '2026-09-04');

INSERT INTO Orders Values
(401, '2026-09-01', 55000, 1, 301, 201, 301),
(402, '2026-09-02', 799, 2, 302, 202, 302);

INSERT INTO Order_item Values
(401, 101, 1)
(402, 102, 2);
```

Output →

```
Select * from Customer;
Select * from Seller;
Select * from Category;
Select * from Product;
Select * from Payment;
Select * from address;
Select * from Orders;
Select * from Order_item;
```

RESULT →

All tables are created successfully and sample records are inserted. foreign - key relationships maintain referential integrity.

09/10/24